

SIEMENS ENERGY · KAZ POWER PLANT UPGRADE PROJECT · KHOR AL-ZUBAIR, BASRAH — IRAQ

MANAGEMENT & STAFF EHS INDUCTION

Rev. 01 — Proposed Full Rewrite (Corrected & Aligned with Siemens Energy EHS Requirements)

Document No. **KAZ-EHS-IND-001** · Rev. **01** · Status: **For Review & Approval**

Client: Ministry of Electricity (MoE) & Basrah Gas Company (BGC) · Principal Contractor: Siemens Energy

Prepared by: EHS Manager (Al-Mial / AMC) · Reviewed by: SE EHSMIP · Approved by: Project Director / EHS Director

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Zero Harm · Zero incidents is achievable · Health and safety – no compromises · We take care of each other

SLIDE 1 — Cover

KAZ POWER PLANT UPGRADE PROJECT

Management & Staff EHS Induction

Khor Al-Zubair Gas Turbine Power Plant — Basrah, Iraq

Client: Ministry of Electricity (MoE) & Basrah Gas Company (BGC) · Principal Contractor: Siemens Energy

Presented by: Al-Mial (AMC) EHS Department | Document No. KAZ-EHS-IND-001 | Rev. 01 | Date: __

Approved by: Project Director / EHS Director / SE EHSMIP

ولا عن تصريح العمل PSI ولا يُغني عن Siemens Energy ملاحظة للمدرب: يُذكر صراحة أن هذا الإندكشن جزء من متطلبات

SLIDE 2 — Emergency instructions for this session

- If an emergency occurs during this induction: stop the session, follow the exit signs and proceed to the nearest **Assembly Point (Muster Point)** — location:
- Alarm signal: continuous siren = evacuate · intermittent = stand by · all-clear = return to work
- Mobile phones on silent (keep your phone available for emergencies)
- Smoking is prohibited except in the designated smoking area (never in red/restricted zones)
- Emergency numbers (see Slide 63): Site EHS Control Room — · UHF Channel 1 · Ambulance 122 · Fire 115 · Police 104 · Siemens Energy Emergency +90 533 625 9494

SLIDE 3 — Contents (must match the actual deck)

- Objectives & validity · 2. Project description & site · 3. HSE policy & SE Zero Harm · 4. Roles & responsibilities · 5. General site rules · 6. Stop Work Authority · 7. Risk management (RA/SJA/LMRA/Toolbox/MoC) · 8. Permit to Work & LOTO · 9. Electrical safety & arc flash · 10. High-risk activities (height, lifting, excavation, confined space, hot work, tools) · 11. PPE · 12. Health, heat stress & first aid · 13. Environment, waste & chemicals · 14. Fire & firefighting · 15. Security, access & journeys · 16. Driving & mobile plant · 17. Emergency response & incident reporting · 18. Lessons learned · 19. Violations & penalties · 20. Assessment, card & sign-off

SLIDE 4 — Objectives

At the end of this induction you will be able to:

- State the project HSE policy and the **Siemens Energy Zero Harm** framework
- Apply the general site rules and use your PPE correctly
- Exercise **Stop Work Authority** and explain your protection from retaliation
- Read a risk assessment, complete an **LMRA** and know when an **SJA** is required
- Explain how the **Permit to Work (PTW)** and **LOTO** systems work on this site
- Recognise the electrical, height, lifting, excavation, confined-space and hot-work hazards of KAZ
- Respond correctly to a fire, an emergency, an injury, a spill or a security event
- Report incidents, near misses, unsafe acts and conditions within the required timeframes

SLIDE 5 — Validity, assessment and records

- Induction validity: **12 months**; renewal on expiry; re-induction after a serious incident or a change of hazards
- Knowledge check: **10 questions — pass mark 80%** (retraining required if below)
- Your attendance and signature are recorded in the site training register
- The induction card must be carried and shown on request — **the card does not replace the Permit to Work**
- Minimum site age: **18 years**

SLIDE 6 — Project description

The KAZ Power Plant Upgrade Project modernises and expands the existing **400/132/11 kV Air Insulated Switchgear (AIS)** facilities at the Khor Al-Zubair Power Plant, Basrah, Iraq.

Scope includes: new **400 kV and 132 kV bays** (circuit breakers, disconnectors, earthing switches, CTs, VTs, surge arresters, busbar extensions), new foundations, earthing grid enhancement, steel structures, and all control, protection, metering, telecom and SCADA/DAS cable works, including civil, mechanical, electrical installation, testing, commissioning and trial operation.

Project duration: 13.5 months · Working hours: see Slide 15

Client: **MoE & BGC** · Principal Contractor: **Siemens Energy** · Subcontractor: **Al-Mial (AMC)**

The works are executed inside a LIVE power plant — coordination with MoE Operations is mandatory for every activity.

SLIDE 7 — Site plot plan, work areas and emergency facilities

[Insert simplified plot plan with bilingual legend]

- Project working area · Mobilisation / laydown area (TSF) · Main gate · Warehouses · Camp & offices
- **Emergency layer:** assembly points, first-aid points, fire points, eyewash, emergency exits, ambulance access route, nearest hospital

في مفتاح الخريطة (TSF ملاحظة للمدرب: يجب تعريف أي اختصار يظهر في المخطط (مثل

SLIDE 8 — Project EHS organisation & contacts

[Organisation chart with names/functions]

- Project Director / Site Manager — overall responsibility for EHS
- **SE EHSMIP / SE EHS Manager** — governance, audits, verification, Stop Work authority
- **EHS Manager (Contractor)** — daily supervision, permits, inspections, training
- **EHS Officers** — one per work area/shift (No EHS supervision = No work)
- **Area Authority / Operations Manager (MoE)** — authorises work in live operational areas
- **AEP / Isolation Authority** — energy isolation, LOTO, energisation
- First aiders, fire wardens, gas testers — posted in every area

SLIDE 9 — HSE policy

Siemens Energy EHS Policy (Annex 1 — SE Standard 000050) and the KAZ Project EHS Plan apply to everyone on this site.

Project commitment: **Zero Harm** — zero injuries, zero occupational illnesses, zero environmental incidents, zero security breaches.

Project target: **Zero H1 = ZERO.**

- Safety takes precedence over progress and cost
- Everyone has a duty to protect themselves, colleagues, the environment and assets
- Health, safety and environmental protection are conditions of employment on this project
- Contractor policy: Al-Mial (AMC) — “**Do it safely, or don’t do it**” (supporting the SE framework)

SLIDE 10 — Siemens Energy Zero Harm framework

- **Zero incidents is achievable · Health and safety – no compromises · We take care of each other**
 - Zero Harm behaviours: take the time to assess risk and plan; talk about safety before and during the work; stop and re-assess when things change; follow the rules; intervene and never walk by
 - Zero Harm Essentials: the Stop principle, safe systems of work, competency, risk assessment, PTW, PPE, emergency preparedness
- (مثلاً: إيقاف رافعة بسبب الريح، أو إيقاف عمل بدون قفل شخصي) KAZ ملاحظة للمدرب: اربط كل مبدأ بمثال من موقع

SLIDE 11 — What Zero Harm means for you today

- Plan your task before you start it — never improvise around a hazard
- Use the right permit for the right job
- Verify your PPE before every task
- Report every near miss, unsafe act and condition — reporting is not blame
- Intervene politely when you see risk — it is expected, not optional

SLIDE 12 — Your responsibilities

Everyone shall: take care of own and colleagues' safety; comply with site rules, permits and instructions; report hazards, incidents and near misses; use PPE correctly; attend training and toolbox talks; never work under the influence of alcohol or drugs; stop unsafe work.

Supervisors / Superintendents additionally: brief the crew, complete LMRA/SJA, verify permits and certifications, control the work area, verify controls before restart, ensure only competent personnel perform the task.

Managers / Staff additionally: provide resources, verify compliance in their areas, lead safety walks, hold contractors accountable, ensure EHS is the first agenda item in every meeting.

SLIDE 13 — Contractor and subcontractor obligations

- Each contractor/subcontractor works to its own EHS system enriched with Siemens Energy, Client and KAZ project requirements
- Every contractor appoints a competent **Contractor EHS Representative**
- Contractors provide: risk assessments, method statements, permits, competency certificates, equipment certificates, monthly man-hours and incident data
- Siemens Energy may **stop any work**, exclude or ban individuals for violations — at the contractor's cost
- Where local/legal requirements differ from SE requirements, **the stricter requirement applies**

SLIDE 14 — General site requirements (1)

- Complete the **site induction** + **PSI (Personal Safety Instruction)** before starting any work
- **No work without a valid Permit to Work**; no work without EHS supervision (**No EHS supervision = No work**)
- Complete an **LMRA** before every task; attend the daily **Toolbox Talk**
- Complete an **SJA** for high-risk tasks (heights, lifting, confined space, live electrical, hot work...)
- Read and understand the risk assessment and method statement for your task — ask if anything is unclear
- Never start work in an area you have not been authorised to enter (red zones, live areas)

SLIDE 15 — General site requirements (2)

- Working hours: **06:00–15:00, 6 days/week** (subcontractor hours approved by SE); maximum 11 hours daytime / 7.5 hours night work, one rest day in every 7 — no fatigue driving or working
- **Minimum PPE at all times on site** (see Slide 43)
- Only competent, certified persons operate plant and equipment
- Keep your workplace **clean and tidy** — “**Clean everything – always!**”
- Report every near miss, unsafe act, unsafe condition and incident (Slide 62)

SLIDE 16 — Electrical safety rules (1)

- Treat every electrical circuit as **live until proven dead**
- Isolate the source before work; **LOTO** to prevent re-energisation
- **Prove the tester before and after testing** — then prove dead
- Only an **Authorized Electrical Person (AEP)** may isolate, de-isolate or energise; each worker applies a **personal lock**
- Never work on live equipment unless formally authorised and only under an **Electrical Hot/Commissioning Permit**
- Use insulated tools suitable for the voltage; verify calibration and condition
- Maintain safe approach distances at all times (Slide 31–32)
- Keep panels and access routes clear; report damaged equipment, cables or covers immediately

Golden Rule: ISOLATE — LOCK OUT — TAG OUT — TEST BEFORE TOUCH

SLIDE 17 — Electrical safety rules (2) — Arc flash and switching

- Arc flash risk assessment exists for this site: respect the **Arc Flash Boundary**, use the specified PPE category, never bypass barriers or interlocks
- Only competent, appointed persons may switch, ground or energise (Operations Manager, HV Safety Supervisor, Switching Supervisor, AEP)
- Work within **30 m of a high-voltage system** requires agreement with the plant owner and physical demarcation
- **Stop all work during thunderstorms** — cranes, lifts, scaffolding and installation work must be suspended (lightning and induced voltage)
- Report any contact with a conductor, any flash and any damaged insulation immediately — do not touch the casualty before isolation

SLIDE 18 — Safe distances (project-specific, metric)

[Table approved by SE EHSMIP and the Client — to be inserted]

Nominal voltage	Minimum approach distance (m)	Arc flash boundary (m)	Notes
11 kV	...	...	Limited / restricted approach per NFPA 70E + CV
132 kV	...	...	Client operational rules apply
400 kV	...	...	Client operational rules apply — highest risk on this site

Rules: no crane, MEWP, ladder, scaffold, excavator or vehicle may enter the protected zone; barriers and signage must remain in place until the work is completed; **look up and look around before moving any plant or load.**

SLIDE 19 — Prohibited items and behaviours

- Alcohol, narcotics and other intoxicating substances — **zero tolerance**; random testing enforced; refusal or a positive result = immediate removal and permanent ban
- Firearms and ammunition — prohibited for all personnel (except authorised security personnel)
- **Smoking — only in designated smoking areas**; never in red zones or near flammables
- Weapons, fighting, harassment, discrimination, theft, sleeping and horseplay — zero tolerance
- Photography and video recording **anywhere on site** — prohibited without written authorisation from the Site EHS/Security Manager
- Mobile phones — not in red zones/restricted areas; never while driving (including hands-free)
- Removal of any material, tool, equipment or waste requires a **Material Gate Pass**

SLIDE 20 — Stop Work Authority

- **Anyone** may stop any task — employee, contractor, visitor — when there is imminent danger, a rule violation, a missing control or a change in conditions
- Steps: 1) Stop the activity · 2) Intervene directly with those involved · 3) Escalate to the supervisor/area authority · 4) Verify and implement controls · 5) Restart only when authorised · 6) Record and feed back
- **You will not be punished, prosecuted or harassed for stopping work** — this is a Siemens Energy commitment
- Work remains stopped until the controls are in place and the area authority authorises a restart
- Siemens Energy reserves the right to stop work and suspend any activity that is unsafe

SLIDE 21 — Risk management: RA, SJA and LMRA

- **Risk assessment (RA):** read it, understand it, follow the controls; never work outside its scope
- **SJA (Safe Job Analysis):** required for high-risk/non-routine tasks; prepared with the crew before work starts
- **LMRA (Last Minute Risk Assessment):** short check before every task — What can hurt me? What has changed? Are controls in place? If in doubt — stop and ask
- **Management of Change:** if the method, equipment, personnel or site conditions change, work stops and the assessment is revised and approved

SLIDE 22 — Daily safety routine

- Start of shift: toolbox talk → LMRA → permit verification → PPE check → equipment pre-use check
- During work: supervision present, housekeeping maintained, barriers and signage in place, no lone working on high-risk tasks
- End of shift: close/endorse permits, restore isolations per the LOTO register, secure the area, hand over outstanding tasks in writing
- Shift handover for isolated work: joint re-verification of permit conditions, isolation status and zero-energy state with signatures

SLIDE 23 — Permit to Work (1) — system and roles

- **No permit = No work.** The PTW system is controlled by Siemens Energy; for work on live/operational plant the Client (MoE/BGC) permit system takes precedence
- Roles: **Permit Issuer (SE) · Permit Holder (contractor supervisor) · Area Authority (operations) · Isolation Authority/AEP · EHS Reviewer · Authorized Gas Tester**
- All permits require: approved JSA/method statement, verified controls on site, briefed working party, valid certificates and equipment inspections
- EHS reviews and audits permits; technical approval belongs to the Area Authority; isolation authority belongs to the AEP

SLIDE 24 — Permit to Work (2) — permit types and validity

Permit	Typical validity
General Cold Work	7 days with daily endorsement
Electrical Cold Work	7 days with daily endorsement
Electrical Hot Work & Commissioning	Daily / each shift
Energy Isolation (LOTO)	Daily / each shift
Hot Work (welding, cutting, grinding)	Daily / each shift
Excavation & Trenching (> 30 cm)	7 days with daily endorsement
Lifting (all lifts; detailed plan for critical lifts)	Daily / each shift
Work at Height (≥ 1.8 m)	Daily / each shift
Confined Space Entry	Daily / each shift

SLIDE 25 — LOTO and energy isolation

- Cover **all** energy sources: electrical, mechanical, pneumatic, hydraulic, thermal, stored energy
- Sequence: notify → isolate → lock & tag → release stored energy → verify **zero energy** → work
- Every worker applies their **own personal lock**; keys are kept in the lockbox under the AEP's control
- **PTW validity** ≠ **LOTO validity** — expiry of a permit does not release an isolation; only the AEP de-isolates through the controlled process
- Never remove another person's lock or tag — this is a dismissible offence

SLIDE 26 — Work at height

- **Fall protection is required from 1.8 m** (or where a fall hazard exists)
- Use approved scaffolding, platforms or MEWPs; harness with **double lanyard and shock absorber**, anchored to a certified point (> 22 kN), **100% tie-off**
- Scaffolding: only erected/modified by certified scaffolders, **green tag** valid, inspected before each shift/after alteration, no work on red-tagged scaffolds
- Ladders: short-duration work only, secured, 3 points of contact, no carrying loads, no working from the top steps
- Barricade the area below; tether tools; never throw materials; protect openings and edges

SLIDE 27 — Lifting operations

- **All lifts require a permit;** critical lifts (load > 75% of capacity, multi-crane, over live plant/energised equipment, non-routine) require a detailed engineered lifting plan
- Requirements: certified crane and lifting tackle (third-party certificates), competent operator, rigger and banksman, exclusion zone, taglines
- Outriggers fully extended on timber pads on firm ground; verify ground bearing capacity
- **No lifting in wind above 32 km/h** or in poor visibility; never lift over personnel or live equipment
- Load control: no side pulls, no shock loading, nobody under the load

SLIDE 28 — Excavation and trenching

- A permit is required for any ground penetration deeper than **30 cm**
- Obtain the signed **underground utility clearance** (electrical/cables/earthing grid) before digging — **hand-dig trial pits near services**
- Protective systems (shoring, benching or sloping) are mandatory for excavations deeper than **1.2 m**
- Spoil heaps set back at least **1.0 m** from the edge; barricade and sign all excavations; ladders for egress every 7.5 m
- Inspect after rain or any ground movement; keep plant and vehicles away from the edge; never enter an unprotected trench

SLIDE 29 — Confined spaces

- Entry only with a **Confined Space Entry Permit**, trained entrants/attendants and a rescue plan
- Atmospheric testing before entry and continuously during entry: **O₂, LEL, H₂S, CO** by an Authorized Gas Tester
- Ventilation, isolation (LOTO) and rescue equipment in place; the attendant stays at the entry at all times
- **Never enter to rescue** without breathing apparatus and the rescue team — most confined-space fatalities are would-be rescuers
- Excavations are classified as confined spaces only by risk assessment, not automatically

SLIDE 30 — Hot work and fire prevention

- **Hot Work Permit** required for welding, cutting, grinding, soldering, thermal spraying or any spark/flame/heat above 100 °C
- Requirements: competent certified welder, fire blanket containment, **10 m clearance** of combustibles, dedicated **Fire Watch during the work and for 30 minutes after**, minimum two extinguishers on site, gas cylinders secured upright, separated (fuel/O₂) and capped
- Notify the area and the fire point before starting; check for hidden combustibles; screens to protect others
- **Never** perform hot work where a flammable atmosphere may exist or near fuel/chemical areas without a gas test

SLIDE 31 — Hand and power tools

- Use the right tool for the job; inspect before use; remove defective tools from service immediately
- Portable electrical tools: 230 V with **30 mA RCD**, armoured/undamaged cables, earthed (verify earthing measurement), no taped joints
- Angle grinders: guard in place, correct disc for the material, no disc over-speed, wear **full face shield (EN 166)** + safety glasses + gloves
- Secure the workpiece; avoid cutting towards your body; no loose clothing, no jewelry, no rings, tie back long hair
- Compressed air is not for cleaning clothes or skin

SLIDE 32 — Manual handling and ergonomics

- Assess the load: weight, size, grip, centre of gravity, distance and destination
- Use mechanical aids or get help; bend knees, keep the load close, no twisting while lifting
- Plan breaks and vary tasks; report any pain early — do not wait for it to become an injury
- Team lifting with a leader and clear commands for heavy or awkward loads

SLIDE 33 — PPE: mandatory minimum on site

- Helmet with **chin strap** + high-visibility vest
- **Flame-retardant coverall** (visibility class 1)
- Safety shoes with toe protection and puncture resistance (**S3/S1P**)
- Safety glasses (ANSI/EN certified)
- Protective gloves (task-specific) and hearing protection in high-noise areas (> **85 dB(A)**)
- Additional PPE is specified by the task risk assessment: arc-rated PPE (Category 2/4), face shield, insulated gloves, full-body harness, chemical-resistant PPE, respiratory protection
- **Respiratory protection is mandatory where dust/vapour is identified (e.g., FFP3) and requires face-fit testing before use**
- PPE must be in good condition and a good fit — damaged PPE is replaced free of charge

SLIDE 34 — Health, heat stress and hydration

- Heat stress is managed by **WBGT monitoring** (monitoring starts when ambient temperature exceeds **40 °C**) with published work/rest cycles per task intensity
- **Acclimatisation** applies to all new arrivals and to anyone returning from leave: reduced workload and increased rest in the first days
- Drink water regularly (**1.0–1.2 litres per hour** for light work in high heat); use oral rehydration salts; shaded, ventilated rest areas are provided
- **Lone working is prohibited** on high-risk tasks — use the buddy system
- Warning signs: headache, dizziness, cramps, nausea, confusion, hot dry skin, fainting → move to shade, cool the body, give fluids if conscious, call the medic/UHF Ch.1 immediately — **heat stroke is a medical emergency**
- Ramadan and summer peak-hour arrangements are published separately and must be followed

SLIDE 35 — First aid and medical emergency

- First-aid points, trained first aiders, eye-wash stations, stretchers and the site clinic locations are marked on the plot plan
- Everyone must know: the nearest first-aid point, the site medic, the emergency number and the route to the nearest hospital
- Rules: do not move a casualty unless there is immediate danger; do not touch an electric-shock casualty before isolation; keep the airway open; start **CPR/AED** if trained and the casualty is not breathing; preserve the incident scene
- Meet the ambulance at the main gate and escort it in; report the emergency to the EHS Control Room immediately
- Fitness for work: valid medical certificate, report any illness/injury/medication that affects your ability to work safely

SLIDE 36 — Environment: our obligations on site

- Prevent pollution — spills, leaks, dust, noise and waste are all reportable events
- **No spill of oil, fuel, chemical or concrete washwater to soil or drainage** — use drip trays, secondary containment, funnels and spill kits
- Dust suppression on unpaved roads and cutting operations; noise and vibration controlled by selection of plant and work methods
- **Asbestos and PCBs are prohibited on site**; suspected material must be reported and treated by certified personnel
- Protect flora, fauna and heritage: no uncontrolled clearing, no hunting, no littering
- Report environmental incidents (leaks, spills, uncontrolled emissions) immediately — they are classified as EHS incidents

SLIDE 37 — Waste management

- Segregate at source: general waste · recyclables (metal, cable, paper, plastic) · **hazardous waste** (oil-contaminated, batteries, filters, chemicals, e-waste) · construction/demolition waste
- Hazardous waste is stored in labelled, covered, bunded areas and moves only with **MOTAT records and a licensed transporter**
- Never burn waste, never mix hazardous with general waste, never dump into trenches, watercourses or empty land
- Material and waste removal requires a **Material Gate Pass**
- Keep all waste areas tidy, marked and away from drains

SLIDE 38 — Chemicals and hazardous substances

- No chemical may be brought to site without a **Safety Data Sheet (SDS)** and registration in the electronic substance register
- Read the label and the SDS; use the specified PPE; never decant into unlabelled containers
- Store flammables separately in approved cabinets, in limited quantities; keep oxidisers away from fuels
- Eye-wash and emergency shower must be accessible where chemicals are handled
- Report all contact, inhalation or ingestion immediately

SLIDE 39 — Fire classes and extinguishers (corrected)

Fire classes (state the standard used):

- **Class A** — wood, paper, cloth, rubber, some plastics → water, foam (DCP acceptable but less effective)
- **Class B** — gasoline, kerosene, paint, propane → foam, DCP, CO₂
- **Class C (NFPA)** — energised electrical equipment → **CO₂ or clean agent** + **de-energise the supply** (never water)
- **Class D** — metals (magnesium, titanium, sodium) → **special metal powder** (graphite/NaCl) — **NEVER water or standard DCP**
- **Class K/F** — cooking oils and fats → **wet chemical** (camp kitchen)

[Images of each extinguisher type]

SLIDE 40 — Using a fire extinguisher (PASS)

- Pull the pin · Aim at the base of the fire · Squeeze the handle · Sweep side to side
- Hold the extinguisher upright; stand between the fire and your escape route, back to the exit
- For CO₂: hold the **insulated horn**, never the metal parts (cold burn risk)
- **Only fight a small fire if you are trained and have a clear escape route** — otherwise raise the alarm and evacuate
- Call the alarm first; evacuate if the fire grows, produces dense smoke or involves gas cylinders
- Extinguishers are inspected monthly and serviced annually — never use a defective or missing extinguisher: report it

SLIDE 41 — Security and access control

- Site and camps are protected by the Iraqi Police and site security; all persons, belongings, vehicles and deliveries may be searched on entry and exit
- Your **Site ID / Gate Pass** must be worn visibly at all times; access without it is refused
- Visitors: pre-approval 24 h in advance, photo ID deposited at the gate, escorted by the host at all times; no manual work, no plant operation and no photography
- Lost or damaged ID card: report immediately (replacement fee applies)
- Security personnel are authorised to carry weapons — cooperate calmly and follow their instructions
- Do not resist or interfere with security checks; theft, unauthorised access, suspicious persons or vehicles must be reported at once
- Red zones (high-voltage areas) are marked — no entry without authorisation

SLIDE 42 — Journey management and site movements

- Movement only with prior authorisation; plan every journey (route, time, weather, checkpoints, contacts)
- Use authorised drivers and vehicles only; no unauthorised passengers; no material transport without a gate pass
- Seat belts for all occupants, at all times, in every vehicle
- Rest at least every two hours; no driving when tired, ill or after long shifts; **no night driving without approval**
- Report any accident, breakdown or security incident on the road immediately
- Emergency contact details must be carried in every vehicle

SLIDE 43 — Driving and mobile plant on site

- Site speed limit **5 km/h**; obey all traffic signs, signals and marshals
- **No mobile phone use while driving — including hands-free**; no eating, no distraction
- Reverse parking only, in designated areas; **reversing requires a trained banksman** — vehicle/plant movements must be controlled by a flagman
- Pre-use checks every day and valid third-party inspection certificates; vehicles display the site vehicle pass, flashing beacon and reversing alarm
- Pedestrians use designated walkways; keep a safe distance from mobile plant; make eye contact with operators
- Driving is restricted in adverse weather (fog, heavy rain, dust storms) — non-essential journeys are postponed

SLIDE 44 — Emergency response & evacuation

Signal	Action
Continuous siren	Stop work, isolate your equipment, proceed to the nearest assembly point
Intermittent siren	Stand by, prepare to evacuate, await instructions
All-clear	Return to work only when authorised

- Steps: raise the alarm → stop the job and isolate energy → proceed to the assembly point → **roll call (head count)** → report any missing person → follow announcements → do not leave or return until instructed
- Fire wardens/area marshals are responsible for sweeping their area and confirming the head count, including visitors and subcontractors
- Do not use vehicles, do not block emergency routes, keep the gate clear for emergency services
- Evacuation for security incidents (threat, intrusion, civil disturbance) follows the same assembly-point principle with security instructions

SLIDE 45 — Emergency classification and notification

Level	Description	Notification
Level 1	Minor injury/health effect, minor damage or minor environmental impact, manageable on site with site resources	Supervisor → EHS → Site Manager (immediate verbal + written within 24 h)
Level 2	Significant event manageable locally with support from the Company or local authorities (major injury, local damage, local impact)	EHS → SE EHSMIP → Project Director + Client (immediate)
Level 3	Major event/crisis beyond the contractor's control, with broader implications, requiring Company/corporate resources	Project Director → SE Corporate/Regional Hub + Client (immediate)

Life-threatening injury, fatality, high-potential incident, major environmental release and security crisis are escalated immediately — never treated as a locally managed event.

Notification: verbal **immediately**, written notification **within 24 hours**, investigation report within **7 days** (unless an extension is agreed), CAPA status reported weekly.

SLIDE 46 — Emergency contacts (posted and verified)

Contact	Number / channel
Site EHS Control Room / emergency line	 (24/7)
UHF radio	Channel 1
Site medic / first aiders	
AMC emergency numbers	0785-285-7404 · 0786-162-9706
Siemens Energy emergency	+ 90 533 625 9494
Ambulance	122
Civil Defence (Fire)	115
Police	104
Nearest hospital / Medevac	

Rule: call the site emergency line first; keep the line free; state your name, company, exact location, what happened, number of casualties and remaining hazards.

SLIDE 47 — Incident, near miss and unsafe condition reporting

- Report **immediately**: all injuries (including first aid), near misses, unsafe acts, unsafe conditions, property damage, environmental releases, security events
- Report through your supervisor and the EHS Department; incidents are recorded in the **Siemens Energy EHS reporting tool (Enablon)**
- Timelines: verbal immediately · written notification within **24 hours** · investigation report within **7 days** · CAPA follow-up weekly
- Investigation: an investigation team is formed by the Contractor, the Company and subcontractors; root causes and corrective/preventive actions are documented
- **Never** withhold information, adjust the scene or delay reporting; reporting is protected — no blame for honest reporting
- Monthly man-hours and incident data must be submitted by every contractor

SLIDE 48 — Learning from incidents

- Lessons Learned and safety alerts are issued after incidents and high-potential near misses
- Each lesson triggers: risk assessment review, toolbox talk, and verification that controls are effective
- Siemens Energy may conduct **Eye on Safety** reviews and its own investigations
- If you are involved in or witness a serious incident: preserve the scene, cooperate fully with the investigation, and provide a written statement

Your experience prevents the next accident — share it.

SLIDE 49 — Violations and penalties

Violation	1st offence	2nd offence	3rd offence / critical
Speeding (> 5 km/h)	Warning + 2-day vehicle ban	1-week vehicle & driver ban	Permanent site ban
PPE non-compliance	Work stoppage + written warning	Contractor fine	ID cancellation + site removal
Red zone / LOTO breach	Work stoppage + 1-week suspension	—	Permanent site expulsion
Lost/damaged ID card	Replacement fee	—	—
Positive alcohol/drugs test	Immediate permanent ban	—	—
Unauthorised material removal	Confiscation + investigation	—	—
Unauthorised photography	Media deletion + warning	—	Permanent site ban

Siemens Energy may stop work, suspend, exclude or ban any individual or contractor for violations of EHS requirements, site rules or the law.

SLIDE 50 — Knowledge check (10 questions)

[Sample questions — insert blank answer sheet]

- What is the site speed limit and the reversing rule?
- What permit is required for work at 2.5 m height?
- When is an excavation classified as requiring a protective system?
- What are the four gases tested before confined-space entry?
- What is the maximum wind speed for lifting operations?
- Which extinguisher for an energised electrical panel fire?
- What are the three reporting deadlines after an incident?
- What must you do before touching any electrical conductor?
- What are the correct emergency numbers in Iraq? 10. What is your right if you see an unsafe act?

Pass mark: 80%.

SLIDE 51 — Induction card and acknowledgement

- Collect your numbered **induction card** — valid **12 months**, must be carried and shown on request
- Sign the attendance and acknowledgement register before leaving this session
- The induction card **does not replace** the Permit to Work, PSI, Safe Start or task-specific training
- Your feedback on this induction is welcome — report content gaps to the EHS Department

Acknowledgement text: “I attended the KAZ Management & Staff EHS Induction, I understood its content, and I accept that compliance with safety rules, permits and Stop Work Authority is a condition of my access to this site.”

Everyone goes home safely — every day.
Siemens Energy Zero Harm · “We take care of each other”
Questions? Contact the EHS Department (Site EHS Control Room — UHF Channel 1)
Thank you

Annexes to be attached to the deck

- Annex 1 — Permit quick-reference card (permit type, issuer, validity, prerequisites).
- Annex 2 — LMRA pocket card (4 questions, stop criteria).
- Annex 3 — Emergency card (numbers, assembly points, hospital route — bilingual).
- Annex 4 — PPE matrix (task vs PPE).
- Annex 5 — Attendance, acknowledgement and assessment form.
- Annex 6 — Arabic/Turkish summary of critical slides (electrical safety, PTW/LOTO, heat stress, emergency, reporting).

Approval block

Role	Name	Signature	Date
Prepared by — EHS Manager (Contractor)			
Reviewed by — SE EHSMIP			
Reviewed by — Construction/Site Manager			
Approved by — Project Director			
Approved by — EHS Director			